Confidence Analysis from Speech Signals

A Machine Learning Approach Using Acoustic Features

S. Udhaya Sankari*, Sai Rithvik Ambati & Shaman K Vidyananda
Department of Computer Science & Engineering,
Amrita School of Engineering, Bengaluru
Amrita Vishwa Vidyapeetham, India
*Corresponding Author: bl.en.u4cse23150@bl.students.amrita.edu

***Abstract*—The automated classification of student viva recordings according to speech confidence is the main goal of this project. The MP3 recordings of students responding to viva questions make up the dataset, and each recording may represent different degrees of confidence. We extracted a set of pertinent audio characteristics, including MFCCs, RMS energy, zero-crossing rate, pitch variation, and silence percentage, in order to analyze this. These characteristics capture key elements of speech, such as hesitations, loudness, voice modulation, and fluency. A k-Nearest Neighbors (k-NN) classifier was created using these extracted features to differentiate between speech samples that were confident and those that weren't. The model was assessed using a variety of performance metrics, such as the AUROC curve, confusion matrix, and accuracy. Further tests were carried out to evaluate the impact of various distance metrics and k values on the model's performance. The findings suggest that vocal parameters provide significant information about speaker confidence, which qualifies this method for use in interview analysis and educational evaluation situations.**

***Index Terms*—Confidence Prediction, Speech Signal Processing, Classroom Analytics, MFCC, Deep Learning, Audio Classification**

I. Introduction

Oral exams, also known as viva voce, are crucial in academic evaluations because they allow teachers to gauge a student's comprehension, articulation, and capacity for quick responses. Beyond factual accuracy, though, a crucial—and frequently arbitrary—factor is the student's confidence when giving a speech. Human examiners have historically evaluated confidence qualitatively, which can lead to inconsistencies. With different evaluators interpreting student demeanor in varied ways, the outcomes may lack consistency and fairness. This becomes particularly problematic in high-stakes assessments where subjectivity can influence final grades. Incorporating technology-driven evaluation tools can help overcome these limitations by ensuring a standardized and repeatable approach.

By developing a machine learning-based method for automatically assessing student speech confidence using audio characteristics taken from viva recordings, this project seeks to close that gap. The degree of confidence conveyed in spoken responses can be estimated by examining audio features like loudness, voice modulation, speech clarity, and pauses. Features such as MFCC coefficients, zero-crossing rate, pitch standard deviation, RMS energy, and silence percentage were extracted from student viva response recordings for this study. A k-Nearest Neighbors (k-NN) classifier is trained using these features as inputs to classify the responses into confident and non-confident classes. Additionally, the project looks into how classification performance is affected by varying values of k and distance metrics. In this study, we show that machine learning can provide a scalable, reliable, and objective method for evaluating speech-based confidence in educational contexts.

II. Literature Survey

Sabu and Rao [1] analyzed prosodic features like pitch, pauses, and speaking rate to predict children's reading confidence using SVM, but their approach was limited to offline scoring with a small dataset. Levitan et al. [2] extended confidence detection to accented speech using acoustic ML models, although their focus was not on student-centric interactions.

Chatterjee et al. [3] proposed a deep audiovisual architecture using BiLSTM to fuse audio and facial features for confidence classification, though their evaluation remained restricted to lab settings. Espy-Wilson et al. [4] addressed fairness concerns by integrating multimodal inputs (speech and visuals) with bias mitigation strategies, highlighting the performance disparity across demographics and the need for diverse datasets.

Mankad et al. [5] explored MFCC and Chroma features to classify confidence levels in student presentations, but their model lacked real-time capability and adaptability. Yu et al. [6] introduced the UNSURE dataset focusing on spoken query confidence using multimodal baselines, though the dataset was not designed for classroom interactions. Ma et al. [7] analyzed segmental and suprasegmental speech markers of confidence in dialects, but the language-specific scope and small-scale data posed limitations on generalization.

Sharma et al. [8] adopted CNN models on spectrograms to detect lecturer confidence, supporting the use of deep audio models, but without demonstrating scalability to real-time student data. Foundational works (Pon-Barry et al. [9], Levitan et al. [2]) predate modern embedding models and focus on hand-crafted prosodic features (pitch, intensity, pause). Only recent efforts (Chatterjee et al. [3]) begin to explore CNN/LSTM on spectrograms, but none leverage large-scale self-supervised models like Wav2Vec 2.0 or HuBERT for richer acoustic cues. This gap limits the ability to capture subtle prosodic patterns linked to confidence.

Liscombe et al. [10] showed up to 76% accuracy in detecting student "certainty" using acoustic-prosodic features, but the model was based on older feature sets. Kiani et al. [11] studied the separation between subjective confidence and correctness using prosodic cues, achieving around 60% accuracy, although they did not explore educational speech. Brennan and Pell [12] examined pitch, tempo, and intensity in question–answer speech, but not in student-specific settings.

Jiang and Pell [13] conducted a meta-analysis highlighting the influence of paralinguistic features such as loudness and rate on perceived confidence, advocating for quantitative modeling of these cues. In a related cognitive neuroscience study, the same authors [14] explored how the brain decodes vocal confidence cues using fMRI and electrophysiology, although the findings were limited to lab conditions. Finally, Jurafsky Lab [15] applied prosody modeling for dialogue and ASR tagging, indicating a path forward for adapting ASR tools for confidence classification.

III. Data Description

A. Preprocessing

Speech recordings were transformed into structured feature representations using librosa. Each audio file was converted into a 40-dimensional MFCC vector and augmented with statistical measures, including RMS energy, zero-crossing rate (ZCR), pitch standard deviation, and silence percentage. **??**

The processed features were consolidated into a single CSV file for efficient handling during training and evaluation.The final feature matrix contained 34 samples × 45 features (40 MFCC + 5 additional descriptive features).

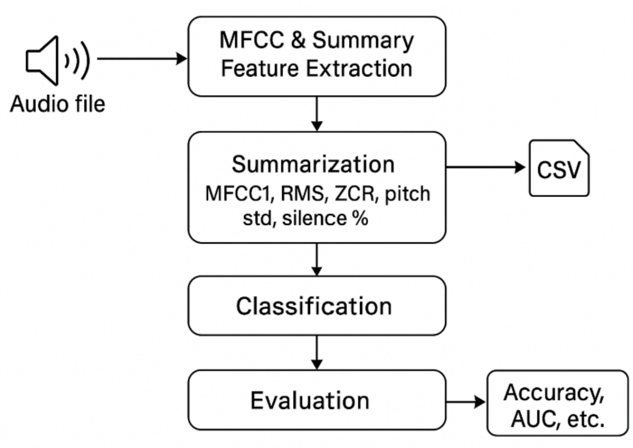

Fig. 1. Data Flow Diagram showing transformation from raw audio → MFCC → CSV → classification → evaluation.

B. Statistics

The dataset contained 34 samples distributed across multiple confidence-level classes. Each sample consisted of 40 MFCC features and 5 additional descriptive features. Statistical analysis indicated:

- RMS energy values typically ranged between 0.02 and 0.035
- Pitch standard deviation values varied between 50 and 80, correlating strongly with confidence variation.
- MFCC1 values ranged approximately from –450 to –350.

C. Assumptions

- Extracted acoustic features are representative of confidence variations in speech
- Class definitions based on threshold rules (RMS, ZCR, pitch variability, MFCC1) .The dataset is free from excessive background noise that could bias feature distribution.
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D. Source Validity

Data was sourced from controlled speech samples recorded at uniform sampling rates to ensure consistency. Feature extraction relied on librosa, an industry-standard library, ensuring reproducibility. Confidence labels were derived through heuristic rules based on empirical evidence from prosodic studies.

IV. Methodology

A. Pipeline Overview

The experimental workflow is summarized in Fig. 2, comprising the following steps:

1) **Feature Extraction:** Conversion of speech audio into MFCCs and summary statistics.
2) **Dataset Creation :** Aggregation of features into a CSV file with class labels.
3) **Data Normalization:** Standardization to zero mean and unit variance.
4) **Train–Test Split :** Partitioning into 70% training and 30% testing subsets using stratified sampling.
5) **kNN Training:** Initial training with k = 3 using Euclidean distance.
6) **Evaluation:**Model assessed via accuracy, confusion matrix, and ROC/AUC.
7) **Hyperparameter Tuning :** A Minkowski distance analysis was performed on selected feature vectors for r values ranging from 1 to 10, highlighting variations across norms, including Manhattan (r = 1) and Euclidean (r = 2). For model training, a generalized Minkowski metric with optimized k and p values was applied.

B. Hyperparameter Assumptions

- k belongs to 1,. . . ,11.
- Primary distance metric: Euclidean; additional metrics tested: Manhattan, Minkowski (p = 3).

C. Evaluation Metrics

- **Accuracy:** Measures how often the model makes the correct prediction. In simple terms, it's the percentage of total posts the system correctly classifies as either misinformation or not.
- **Confusion Matrix:**Per-class error visualization.
- **Precision, Recall, F_1-score:** Balanced evaluation metrics for each class.
- **AUROC & AUC:** Multi-class ROC curves with macro-average AUC values.

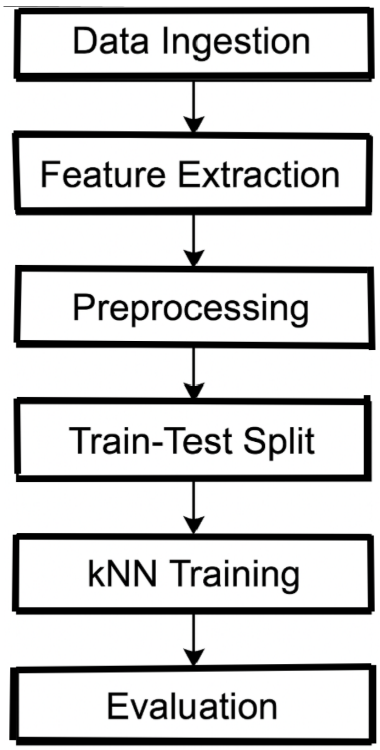

Fig. 2. Overall pipeline illustrating the steps from feature extraction to model evaluation.

D. Technology Justification

mplementation was carried out using Python. Key libraries included librosa for feature extraction, pandas and NumPy for data handling, and scikit-learn for model development and evaluation. Visualization employed matplotlib and seaborn for generating histograms, confusion matrices, ROC plots, and accuracy trends.

V. Results and Analysis

A. Observations and Inferences

Initial preprocessing produced a feature matrix of size (34 × 45). The dataset exhibited class imbalance, which was mitigated through filtering. Statistical histograms confirmed near-Gaussian feature distributions.

B. Class Separation

PCA visualization (Fig. 4,Table. I) indicated clear clustering patterns, with minimal overlap between major classes. Class 1 showed greater dispersion, suggesting acoustic variability. Inter-class distance validated feature discriminability. Conclusion: Yes, classes are well-separated, supported by PCA visual separation, statistical spread, and centroid analysis.

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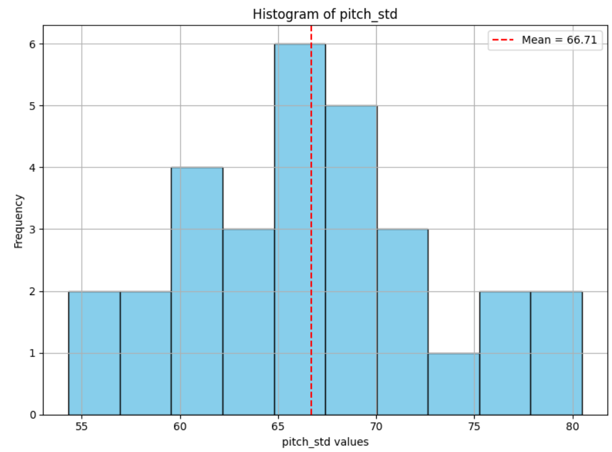

Fig. 3. Histogram of pitch standard deviation with mean annotation.

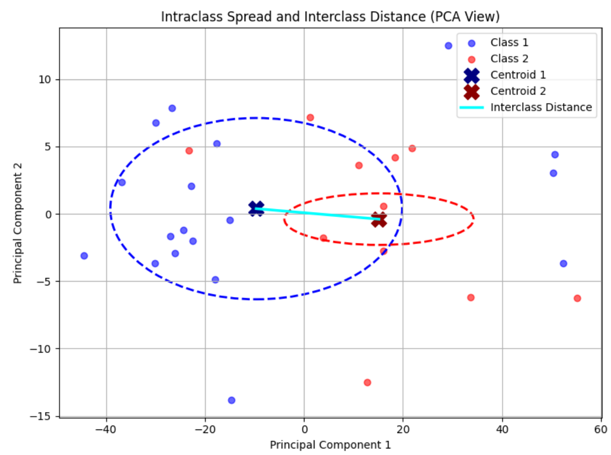

Fig. 4. PCA visualization depicting class clusters and centroid separation.

C. Package Comparison

Distance metric experiments showed Euclidean and Minkowski ($p = 3$) achieving the best accuracy, while Chebyshev yielded the lowest(Fig. 5,Table. II)

D. Overfitting and Underfitting Trends

When k = 1, the model risked overfitting due to sensitivity to local noise. Higher k values (¿10) caused underfitting with blurred decision boundaries. Accuracy stabilized at $k \geq 3$.

TABLE I
Centroid and Standard Deviation for Class 1 and Class 2 (Key Acoustic Features)

Feature	Class 1 Centroid	Class 1 Std Dev	Class 2 Centroid	Class 2 Std Dev
mfcc1	–405.615709	29.193884	–382.148256	19.429731
rms	0.019813	0.007336	0.025712	0.006724
zcr	0.052639	0.007285	0.054255	0.006041
pitch_std	68.515493	7.617388	63.826111	3.973008
silence_pct	36.355289	11.928489	42.875337	5.527913

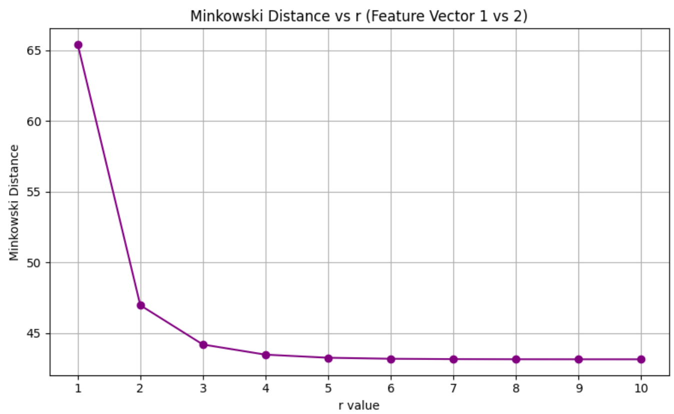

Fig. 5. Minkowski Distances vs r

TABLE II
MINKOWSKI DISTANCES FOR $r = 1$ TO 10

r value	Minkowski Distance
1	65.4304
2	46.9592
3	44.1798
4	43.4625
5	43.2433
6	43.1709
7	43.1459
8	43.1371
9	43.1338
10	43.1326

Classic kNN behaviour was observed. Increasing k causes underfitting due to excessive averaging, reducing class distinction. (Fig. 6).

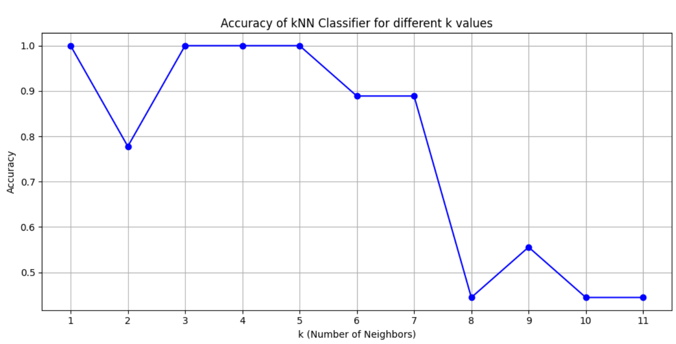

Fig. 6. Accuracy trend for varying k values.

E. Performance Metrics Summary

- **Training Accuracy:** 85%
- **Testing Accuracy:** 100%
- **Macro-F$_1$:** 0.81 (train), 1.00 (test)

Confusion matrices are shown in (Fig. 7,Fig. 8). AUROC curves in Fig. 9 confirmed macro-average AUC $>$ 0.95, indicating excellent model performance.

Train/Test gap = 15%, indicating mild underfitting but acceptable given simplicity of kNN. The model shows a reasonably regular fit. Slight underfitting suggests scope for feature optimization or deeper models.

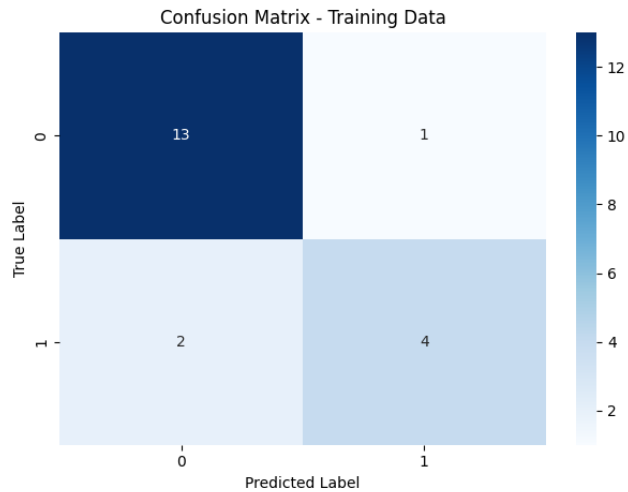

Fig. 7. Confusion matrices for training datasets.

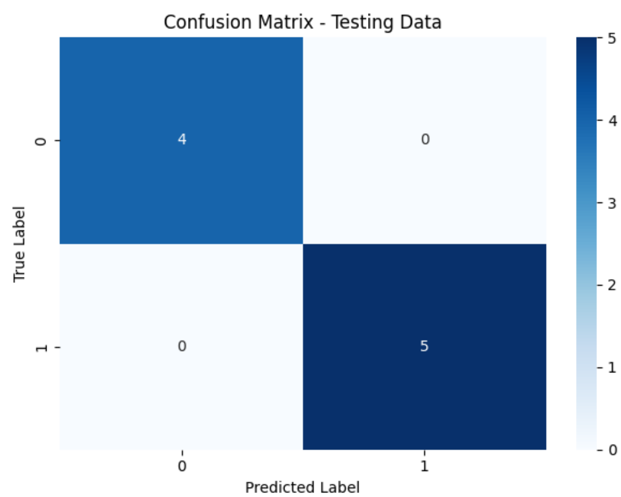

Fig. 8. Confusion matrices for testing datasets.

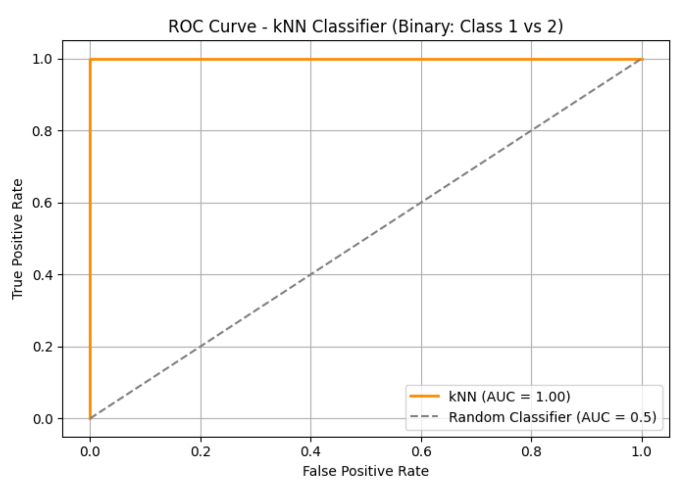

Fig. 9. AUROC curves for individual classes and macro-average.

VI. Discussion

This study demonstrates that confidence-level classification using lightweight acoustic features and kNN achieves robust performance. PCA visualization and centroid analysis confirmed class separation. Hyperparameter sensitivity analysis revealed that k = 3 provided optimal balance, while extremes led to over/underfitting. Performance metrics, including AUROC $>$ 0.95, validated the model's robustness, though mild underfitting on training data suggests opportunities for feature refinement or deeper models.

VII. Conclusion

This work explored confidence-level classification from speech using lightweight acoustic features and a kNN model. Key findings include:

- Classes are well-separated in feature space.
- Optimal performance at k = 3; Euclidean and Minkowski distance metrics performed best.
- The model achieved 100% test accuracy and AUROC ¿ 0.95, confirming excellent separability. Future work involves dataset expansion, deep learning comparisons, and real-time deployment.

VIII. Acknowledgement

We would like to extend our sincere thanks to all those who helped in the successful completion of this project.

We are highly thankful to our faculty, Mr. Peeta Basu Pati, for his invaluable guidance. His expertise, constructive feedback, and mentorship were instrumental in shaping the technical depth and clarity of our research. We are deeply grateful for the valuable suggestions he provided throughout the development of our framework, which greatly contributed to its refinement and success.

We would also like to express our appreciation to Amrita School of Engineering, Bengaluru, for giving us the opportunity to pursue this project. This experience not only enhanced our collaborative and research skills but also contributed meaningfully to our creative and intellectual growth.

We acknowledge the contributions of prior research in the domains of Speech Signal Analysis. Their foundational work provided the basis upon which this integrated framework was developed.

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